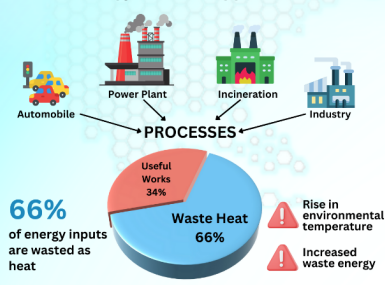


INTRODUCTION

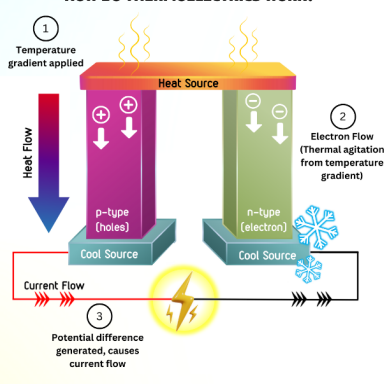
CURRENT LANDSCAPE



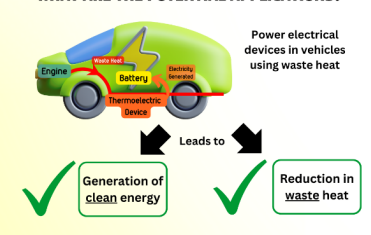
CURRENT SOLUTION

Using THERMOELECTRICS to convert heat into electricity and vice versa

HOW DO THERMOELECTRICS WORK?



WHAT ARE THE POTENTIAL APPLICATIONS?



RESEARCH CHALLENGE

Figure of Merit (zT): Measures efficiency of thermoelectrics

$$zT = \frac{S^2 \sigma}{\kappa} T$$

- S : Seebeck coefficient (V/K)
- σ : Electrical conductivity (S/m)
- κ : Thermal conductivity (W/m.K)

Examples of inter-relations:

- High $\sigma \rightarrow$ Low $R \rightarrow$ Low $V \rightarrow$ Low S
- High $\sigma \rightarrow$ High κ

Increasing zT is a challenge due to inter-relation between S , σ and κ



PROJECT NOVELTY

Problem with current thermoelectric materials: zT is **low** (Due to inter-relation of parameters)

Thermoelectric material: AgSbTe_2

- Very low thermal conductivity (κ)
- Room for further optimization

Limitation with AgSbTe_2 : Suffers from atomic disorder $\rightarrow$ Low electrical conductivity (σ)

PROJECT NOVELTY: GERMANIUM (Ge) DOPING Into Sb

Reasons:

- Ge^{2+} vs Sb^{3+} : Lack of electrons $\rightarrow$ equivalent to holes (Charge carriers) $\rightarrow$ Increase in σ
- Electronegativity of Ge is similar to Sb $\rightarrow$ Fit into Sb sites

Vary amount of Ge (6 samples with 0%, 3%, 6%, 9%, 12%, 15% Ge) in AgSbTe_2

Investigate effect on zT

Properties Measurement

The electrical conductivity (σ) and Seebeck coefficient (S) of all samples were measured with electrical measurement apparatus, and the thermal conductivity (κ) of all samples were measured with laser flash analysis. Then, zT is calculated.

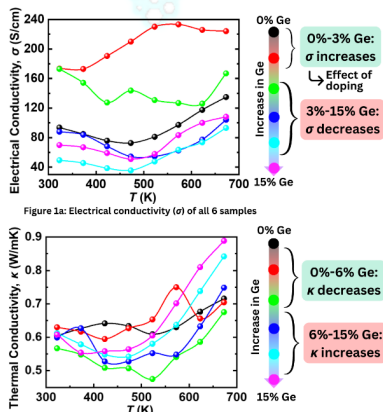


Figure 1a: Electrical conductivity (σ) of all 6 samples

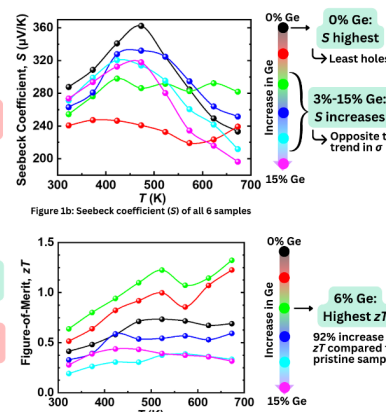


Figure 1b: Seebeck coefficient (S) of all 6 samples

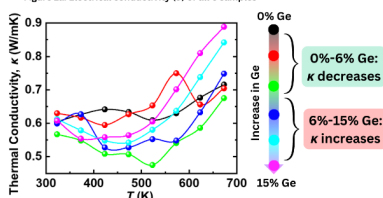


Figure 1c: Thermal conductivity (κ) of all 6 samples

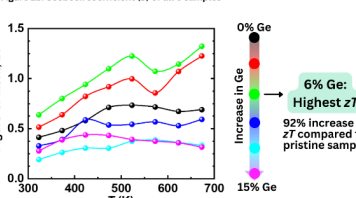


Figure 2: Figure of Merit (zT) of all 6 samples

Analysis:

To investigate the trends observed above, the best sample (6% Ge) was scanned using SEM, and all Ge doped samples were analyzed using XRD

SCANNING ELECTRON MICROSCOPE (SEM)

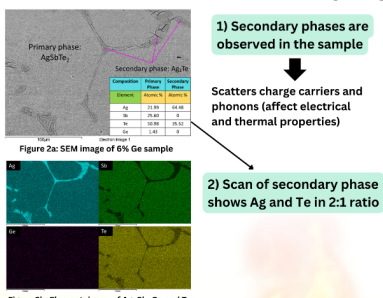


Figure 2a: SEM image of 6% Ge sample

1) Secondary phases are observed in the sample

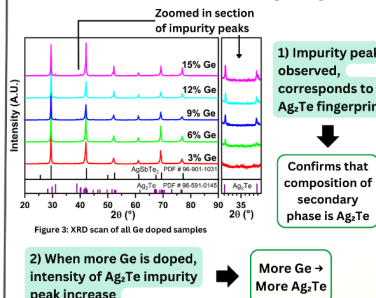
Scatters charge carriers and phonons (affect electrical and thermal properties)

2) Scan of secondary phase shows Ag and Te in 2:1 ratio

Figure 2b: Elemental map of Ag, Sb, Ge and Te

Figure 2b: Elemental map of Ag, Sb, Ge and Te

X-RAY DIFFRACTION (XRD)



2) When more Ge is doped, intensity of Ag_2Te impurity peak increase

More Ge $\rightarrow$ More Ag_2Te

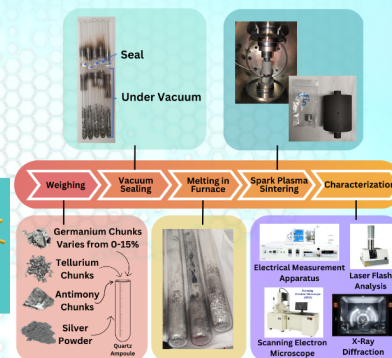
How do these results explain our findings?

For highest electrical conductivity: Optimal trade-offs between effect of doping and charge scattering

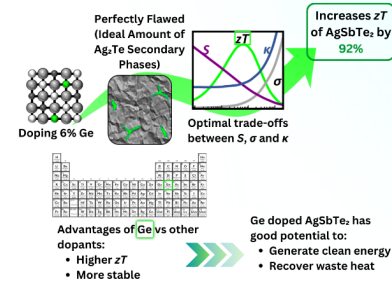
For lowest thermal conductivity: Ideal amount of secondary phases for effective phonon scattering

For highest zT : PERFECTLY FLAWED

EXPERIMENTAL METHODS



CONCLUSION

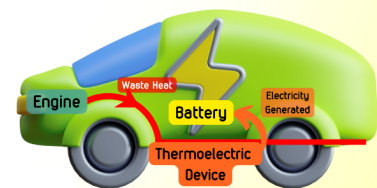


FUTURE WORK

- Exploring Ge as a dopant for other thermoelectric materials
- Further optimization of Ge doped AgSbTe_2 by exploring other synthesis methods (eg: slow cooling)

Improve cost-efficiency and sustainability of thermoelectrics

Realizing potential of thermoelectrics for applications in waste heat recovery



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All photos are self-taken

All figures and diagrams are self-made using PowerPoint
All visual elements are taken from Canva